

Moment
Gen
Funct.

$$\mathbb{E}[e^{\lambda X}] = \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(\lambda X)^k}{k!}\right] = \sum_{k=0}^{\infty} \frac{\mathbb{E}(X^k) \lambda^k}{k!}$$

$$e^a = \sum_{k=0}^{\infty} \frac{a^k}{k!}$$

$$\rightarrow \inf_{k \in \mathbb{N}} \frac{\mathbb{E}(X^k)}{\delta^k} = \frac{\mathbb{E}(X^{k^*})}{\delta^{k^*}}$$

$$\left. \frac{d^n e^a}{da^n} \right|_{a=0} = 1$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\forall k \rightarrow \frac{\mathbb{E}(X^k)}{\delta^k} \geq \frac{\mathbb{E}(X^{k^*})}{\delta^{k^*}}$$

$$\Rightarrow \forall k \rightarrow \mathbb{E}(X^k) \geq \frac{\delta^k}{\delta^{k^*}} \mathbb{E}(X^{k^*})$$

$$\mathbb{E}[e^{\lambda X}] = \sum_{k=0}^{\infty} \frac{\mathbb{E}(X^k) \lambda^k}{k!} \geq \sum_{k=0}^{\infty} \frac{\delta^k \lambda^k}{\delta^{k^*} k!} \mathbb{E}(X^{k^*})$$

$$= \frac{\mathbb{E}(X^{k^*})}{\delta^{k^*}} \sum_{k=0}^{\infty} \frac{(\delta \lambda)^k}{k!}$$

$$\Rightarrow \frac{\mathbb{E}(e^{\lambda X})}{e^{\lambda \delta}} \geq \frac{\mathbb{E}(X^{k^*})}{\delta^{k^*}}$$

$$\rightarrow \inf_{\lambda > 0} \frac{\mathbb{E}(e^{\lambda X})}{e^{\lambda \delta}} \geq \inf_{k \in \mathbb{N}} \frac{\mathbb{E}(X^k)}{\delta^k}$$

□

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0) x^n}{n!}$$

$$\forall \lambda > 0, \mathbb{E}(e^{\lambda(X-\mu)}) \leq e^{\frac{\lambda^2 \sigma^2}{2}}$$

$$\rightarrow \log \mathbb{E}(e^{\lambda X}) - \lambda \mu \leq \frac{\lambda^2 \sigma^2}{2}$$

$$\exists \xi \in [0, x]$$

$$= \frac{f(0) x^0}{0!} + \frac{f'(0) x^1}{1!} + \frac{f''(\xi) x^2}{2!}$$

$$\psi(\lambda) = \log \mathbb{E}(e^{\lambda X})$$

$$- \mathbb{E}^2(X e^{\lambda X})$$

$$\psi''(\lambda) = \frac{d}{d\lambda} \left[\frac{\mathbb{E}(X e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} \right] = \frac{\mathbb{E}[X^2 e^{\lambda X}] \mathbb{E}(e^{\lambda X}) - \mathbb{E}^2(X e^{\lambda X})}{\mathbb{E}^2(e^{\lambda X})}$$

$$= \frac{\mathbb{E}[X^2 e^{\lambda X}]}{\mathbb{E}(e^{\lambda X})} - \left(\frac{\mathbb{E}(X e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} \right)^2$$

$$\forall \lambda = \underbrace{\mathbb{E}[X^2]}_{\lambda} - \underbrace{\mathbb{E}^2[X]}_{\lambda} \leq \frac{1}{4} (b-a)^2$$

upper bound

$$\text{Var}[X] = \mathbb{E}(X^2) - \mathbb{E}^2(X)$$

$$= \mathbb{E}((X - \mu)^2) = \mathbb{E}(X^2 + \mu^2 - 2\mu X)$$

$$= \mathbb{E}(X^2) + \mathbb{E}^2(X) - 2 \mathbb{E}(X) \mathbb{E}(X)$$

$$= \mathbb{E}(X^2) - \mathbb{E}^2(X)$$

$$\mathbb{E}_\lambda[X^2] = \frac{\int_a^b x^2 e^{\lambda x} g(x) dx}{\int_a^b e^{\lambda x} g(x) dx}$$

density of X

$$\begin{aligned} \underline{E_\lambda[X^2]} &= \frac{E[X^2 e^{\lambda X}]}{\rightarrow E[e^{\lambda X}]} \\ &\stackrel{h}{=} \int_a^b x^2 \underbrace{\frac{e^{\lambda x} g(x)}{\int_a^b e^{\lambda y} g(y) dy}}_{h(x) \geq 0} dx \end{aligned}$$

$$\begin{aligned} &\int_a^b h(x) dx \\ &= \int_a^b \frac{e^{\lambda x} g(x)}{\int_a^b e^{\lambda y} g(y) dy} dx \\ &= \frac{1}{c} \int_a^b e^{\lambda x} g(x) dx \\ &= 1 \end{aligned}$$

For a bounded R.V. X , how to find an upper bound on its variance?

$$\text{Var}[X] \leq \frac{(b-a)^2}{4} \quad \checkmark \quad (\text{claim})$$

$$g(t) := E((X - t)^2)$$

$$g'(t) = E(-2(X - t)) = -2E(X) + 2t = 0$$

$$t^* = E(X) \quad g''(t) = 2 > 0 \quad \searrow \uparrow$$

$$\forall t \quad g(t) \geq g(E(X))$$

$$\frac{1}{4}(b-a)^2 \geq g\left(\frac{a+b}{2}\right) \geq \underbrace{g(E(X))}_{\text{Var}[X]}$$

$$g\left(\frac{a+b}{2}\right) = \mathbb{E}\left[\left(\frac{X}{\frac{x}{2} + \frac{x}{2}} - \frac{a+b}{2}\right)^2\right]$$

$$= \mathbb{E}\left[\frac{1}{4} \left(\underbrace{(X-a)}_{\geq 0} + \underbrace{(X-b)}_{\leq 0}\right)^2\right]$$

$$\leq \mathbb{E}\left[\frac{1}{4} \left(\cancel{(X-a)} - \cancel{(X-b)}\right)^2\right]$$

$$= \frac{1}{4} (b-a)^2$$

$$c) \quad \mathbb{E}(e^{\lambda(X-\mu)}) \leq e^{\lambda^2 \sigma^2 / 2}$$

$$\log \mathbb{E}(e^{\lambda X} \cdot \underline{e^{-\lambda \mu}}) \leq \lambda^2 \sigma^2 / 2$$

$$\Rightarrow \log e^{-\lambda \mu} \cdot \mathbb{E}(e^{\lambda X}) \leq \lambda^2 \sigma^2 / 2$$

$$-\lambda \mu + \log \mathbb{E}(e^{\lambda X}) \leq \lambda^2 \sigma^2 / 2$$

$$\log \mathbb{E}(e^{\lambda X}) \leq \underline{\lambda \mu} + \lambda^2 \sigma^2 / 2$$

$$\underbrace{\psi(\xi)}_{\leftarrow} + \frac{\underbrace{\psi'(\cdot)}_{\mu} \lambda}{1!} + \frac{\psi''(\xi) \lambda^2}{2!}$$

$$\underline{\psi''(\xi)} = \text{Var} \leq \frac{(b-a)^2}{4}$$

$$\Rightarrow \log \mathbb{E}(e^{\lambda X}) \leq \mu \lambda + \frac{\frac{(b-a)^2}{4} \lambda^2}{2}$$

$$\Rightarrow \sigma^2 := \frac{(b-a)^2}{4} \Rightarrow \sigma = \frac{b-a}{2} \quad \square$$

